

Supplementary Material B: Control Condition Materials

Content Outline, Session Format, Assessments, and Treatment-Differentiation Verification

Note. This document was compiled from the standardized content guide issued to both the experimental facilitators and the control instructor, from the control instructor’s session records, and from the observer verification notes (August 2026). It is anonymized for peer review. The experimental condition is described in Supplementary Material A.

B1. Overview

The control condition covered the **same AI-fundamentals content** as the experimental condition over the same 10-week schedule: 20 sessions of 90 minutes, twice per week, totaling 30 contact hours. Both conditions received an identical standardized content guide specifying topics, readings, and learning objectives week by week; the conditions differed exclusively in pedagogical approach.

Instruction was delivered by a single instructor with qualifications comparable to those of the experimental facilitators (Electronic Engineering graduate with equivalent post-graduation experience). The instructor was *not* trained in the connectivist framework and did not attend the experimental facilitators’ weekly supervision meetings. The instructor was provided with the same content guide received by the experimental facilitators and was informed that the study was evaluating two instructional approaches for AI literacy.

B2. Session Format

Control sessions followed a consistent three-part format:

1. **Introduction** (10–15 min). The instructor reviewed the previous session’s key points and introduced the session topic via a brief overview slide.
2. **Instructor presentation** (55–65 min). Content was delivered through instructor-led slides and verbal explanation. Students were invited to ask clarifying questions at any point. The instructor demonstrated AI tools (ChatGPT, Google Colab) when relevant to the topic, with students following on their own devices at their discretion; no structured task was assigned during this time.
3. **Closing and reading assignment** (10–15 min). The instructor summarized the session, assigned the individual reading for the following session, and clarified any homework requirements.

No group activities, peer-assessment, ethical-debate structure, or portfolio tasks were included. Student participation was individual and voluntary (questions, note-taking). Seating was in a standard lecture arrangement; students were not organized into fixed teams.

Table B1: Comparison of conditions. Both shared identical content, schedule, contact hours, and tool access; they differed only on the elements marked as “Experimental only”.

Element	Control condition	Experimental condition
Content topics	Same 10-week sequence	Same 10-week sequence
Contact hours	20 sessions $\times$ 90 min (30 h)	20 sessions $\times$ 90 min (30 h)
AI tool access	ChatGPT, Google Colab (incidental)	ChatGPT, Google Colab, Kaggle, GitHub (structured)
Session format	Instructor-led lecture	5-phase connectivist protocol
Group work	Not structured	Multidisciplinary teams of 4–6
Peer teaching	Absent	Weeks 5–7
Ethical debate	Absent	Weeks 1–4 (structured)
Portfolio	Absent	10 weekly entries + integrative reflection
Assessment	2 exams + 4 individual homework tasks	Portfolio (40%) + team project (35%) + peer assessment (15%) + knowledge-node cards (10%)
GitHub use	None	Mandatory commit each session

B3. Weekly Content Outline

The content sequence below follows the same standardized content guide used in the experimental condition (Supplementary Material A, Table A3). Readings and AI-literacy dimensions targeted are the same; session titles and emphases reflect each condition’s instructional format.

Table B2: Control condition weekly content outline, from the same standardized content guide as the experimental condition; readings are identical. Dimensions D1–D4 indicate the AI-literacy domains covered.

Wk	Topic	Session objectives	Reading assigned	Dim.
1	What is AI? Concepts and everyday applications	Distinguish AI, ML, and deep learning. Identify AI applications in students’ own fields. Recognize the gap between AI capabilities and common misconceptions.	Ng & Soo (2024) <i>AI for Everyone</i> , Week 1 notes	D1, D3
2	How do AI systems learn? From data to models	Explain supervised, unsupervised, and reinforcement learning at a conceptual level. Connect data quality to model behavior.	Géron (2022) <i>Hands-On ML</i> , Ch. 1 (conceptual sections)	D1, D2

(Table B2 continued)

Wk	Topic	Session objectives	Reading assigned	Dim.
3	Large language models: architecture and capabilities	Describe transformer architecture at a non-mathematical level. Explain tokenization, context window, and hallucination.	Anthropic (2024) <i>AI Fluency</i> : Description and Discernment modules	D1, D2
4	AI tools in practice: Google Colab and Python for data analysis	Navigate Google Colab independently. Run a basic data-analysis pipeline (load, clean, visualize).	Kaggle (2024) <i>Pandas tutorial</i> , notebooks 1–3	D2, D4
5	AI and bias: sources, mechanisms, and mitigation	Identify three structural sources of algorithmic bias. Apply a bias-evaluation lens to a real system.	Barocas et al. (2023) <i>Fairness and ML</i> , Ch. 1–2 (selected)	D3, D4
6	Collaborative AI projects: datasets and problem framing	Understand the workflow of a data science project. Evaluate a dataset for relevance, quality, and ethical provenance.	Kaggle (2024) <i>Data documentation standards</i>	D2, D4
7	AI-generated content: risks, rights, and disciplinary implications	Distinguish categories of AI-generated content. Analyze copyright and attribution issues.	UNESCO (2023) <i>Guidance for Generative AI in Education</i> , Sections 2–3	D3, D4
8	Building and evaluating an AI project	Understand the components of an end-to-end AI pipeline. Apply evaluation criteria to an AI output.	Instructor-prepared case summary (two pages)	D2, D4
9	AI and society: labour, inequality, and access	Analyze AI's differential societal impact across socioeconomic and geographic contexts. Connect abstract principles to local cases.	Valdivieso (2025) on AI access in Latin America (selected); Torres-Díaz et al. (2025) on AI and academic integrity	D1, D3, D4

(Table B2 continued)

Wk	Topic	Session objectives	Reading assigned	Dim.
10	Integration: what AI can and cannot do — current state and future trajectories	Synthesise learning across the four AI-literacy dimensions. Identify open questions for continued learning.	No new reading (review of previous materials)	D1, D2, D3, D4

B4. Assessment

The control condition used two written examinations and four individual homework assignments, following a conventional assessment model. No portfolio, peer-assessment, or team project was included. Table B3 summarises weights and timing.

Table B3: Control condition assessment components.

Component	Weight	Timing	Format
Written examination 1 (mid-term)	30%	End of Week 5	Written, in-class; 60 minutes; 20 multiple-choice items (MCQ) + 2 short-answer questions
Written examination 2 (final)	40%	End of Week 10	Written, in-class; 90 minutes; 30 MCQ + 3 short-answer questions
Individual homework assignments (4)	30%	Weeks 2, 4, 6, 8	Individual written submission; 300–500 words each; submitted via LMS

B4.1. Written Examination 1 — Sample (Mid-term, Week 5)

Instructions to students: Answer all questions. Multiple-choice questions have one correct answer. Short-answer questions should be answered in 80–120 words each. Time allowed: 60 minutes. No AI tools permitted during the examination.

Part I: Multiple-Choice Questions (20 items, 1 point each)

The following are representative items drawn from the examination bank used in Week 5. Items are organized by the four AI-literacy dimensions.

Conceptual Understanding (D1) — Items 1–6

- Which of the following best describes the relationship between Artificial Intelligence, Machine Learning, and Deep Learning?
 - They are three independent and unrelated fields of computer science.
 - Machine Learning is a subset of AI; Deep Learning is a subset of Machine Learning.
 - Deep Learning is the broadest field; AI is a specialised subfield of Deep Learning.

- d) Machine Learning and Deep Learning are interchangeable terms for the same approach.
- 2. A model trained to classify emails as spam or not-spam, using a labeled dataset, is an example of:
 - a) Unsupervised learning b) Reinforcement learning c) Supervised learning d) Rule-based programming
- 3. The term “hallucination” in the context of large language models refers to:
 - a) The model generating visually realistic images from text.
 - b) The model producing plausible-sounding but factually incorrect or unverifiable statements.
 - c) The model refusing to answer certain categories of questions.
 - d) The model retrieving documents from an external database.
- 4. In a transformer-based language model, the “context window” refers to:
 - a) The user interface through which prompts are submitted.
 - b) The maximum amount of text (in tokens) the model can process in a single interaction.
 - c) The internal memory that persists across separate conversations.
 - d) The number of layers in the neural network architecture.
- 5. Which learning paradigm requires no labeled data and seeks to discover hidden structure within a dataset?
 - a) Supervised learning b) Reinforcement learning c) Transfer learning d) Unsupervised learning
- 6. A recommendation system that learns which content to show a user by receiving positive or negative signals based on user engagement is best described as:
 - a) Supervised learning b) Deep learning c) Reinforcement learning d) Unsupervised learning

Technical Skills (D2) — Items 7–10

- 7. In a Python data-analysis workflow using `pandas`, which operation is performed by `df.dropna()`?
 - a) It removes duplicate rows from the DataFrame.
 - b) It removes all columns with at least one missing value.
 - c) It removes all rows that contain at least one missing value.
 - d) It replaces missing values with zero.
- 8. A confusion matrix for a binary classifier shows: True Positives = 80, False Positives = 20, True Negatives = 90, False Negatives = 10. What is the precision of the model?
 - a) 0.80 b) 0.89 c) 0.75 d) 0.90
- 9. When using ChatGPT for a complex analytical task, which prompting strategy asks the model to reason through intermediate steps before providing a final answer?
 - a) Zero-shot prompting b) One-shot prompting c) Chain-of-thought prompting d) Temperature scaling
- 10. In the context of the GAME framework for AI agents, the “Environment” component refers to:
 - a) The programming language in which the agent is implemented.
 - b) The external systems, data sources, or contexts with which the agent can interact.
 - c) The set of rules that prevent the agent from taking harmful actions.
 - d) The user’s natural language prompt.

Ethics and Social Impact (D3) — Items 11–16

- 11. Historical bias in an AI hiring tool most commonly arises from:
 - a) The use of too many features in the model.

- b) Training on historical decisions that reflected past discriminatory patterns.
 - c) The absence of a training dataset.
 - d) Using a supervised rather than unsupervised learning approach.
12. A medical AI system achieves 95% overall accuracy but performs significantly worse for underrepresented demographic groups. This is an example of:
- a) Overfitting b) Representation bias c) Hallucination d) Regularisation failure
13. Under what condition is it ethically unproblematic to use a large publicly available dataset to train a commercial AI model?
- a) When the dataset contains more than one million records.
 - b) When data subjects consented to their data being used for commercial AI training.
 - c) When the model achieves high accuracy on a test set.
 - d) When the training is completed within a secure server.
14. The Structured Academic Controversy (SAC) method for ethical debate differs from a standard debate in that:
- a) Participants must always defend the position they personally believe.
 - b) Participants must argue both sides before reaching a negotiated consensus.
 - c) The instructor assigns a winning team at the end.
 - d) Participants are not allowed to change their position once stated.
15. Which principle from UNESCO’s guidance on generative AI in education most directly addresses the risk of deepfake academic submissions?
- a) Sustainability b) Transparency and explainability c) Inclusion d) Privacy
16. Copyright protection for AI-generated text currently:
- a) Is identical to copyright for human-authored text in most jurisdictions.
 - b) Is not granted in most jurisdictions because copyright requires human authorship.
 - c) Is automatically assigned to the AI system developer.
 - d) Is governed solely by the terms of service of the AI platform used.

Collaboration and Knowledge Transfer (D4) — Items 17–20

17. In the context of a data science project team, “problem framing” refers to:
- a) Selecting the programming language to be used.
 - b) Translating a real-world need into a concrete, tractable analytical question that specifies the target variable and success criteria.
 - c) Writing the final project report.
 - d) Splitting the dataset into training and test sets.
18. Peer teaching in interdisciplinary AI teams is most effective when:
- a) Only students with the highest grades teach the others.
 - b) Students with stronger technical backgrounds explain concepts to peers who bring complementary disciplinary knowledge, enabling bidirectional learning.
 - c) Teaching roles are assigned permanently for the duration of the course.
 - d) The instructor monitors all peer interactions and corrects errors immediately.
19. The primary purpose of version control (e.g., Git) in a collaborative AI project is to:
- a) Automatically optimise model hyperparameters.
 - b) Track changes to code and documents over time, enabling collaboration, rollback, and auditability.
 - c) Encrypt sensitive training data.

- d) Deploy the finished model to a production server.
- 20. The “knowledge transfer” dimension of AI literacy refers to a student’s ability to:
 - a) Download a pre-trained model from a public repository.
 - b) Apply AI concepts and tools learned in one domain to problems in a different disciplinary or professional context.
 - c) Transfer data between platforms without format errors.
 - d) Explain the mathematical foundations of neural networks to an expert audience.

Part II: Short-Answer Questions (2 items, 10 points each)

- 21. *A retail company uses an AI model to predict which customers are likely to default on credit payments. The model was trained on five years of historical transaction data from an urban customer base. Describe two sources of bias that may affect this model, explain the mechanism through which each bias arises, and propose one mitigation measure for each. (80–120 words)*
- 22. *Explain the difference between a large language model (LLM) and a traditional rule-based chatbot. In your response, identify one capability that LLMs have that rule-based systems lack, and one risk that arises specifically from that capability. (80–120 words)*

B4.2. Written Examination 2 — Sample (Final, Week 10)

Instructions to students: Answer all questions. Time allowed: 90 minutes. No AI tools permitted. Part I: 30 MCQ (1 point each). Part II: 3 short-answer questions (10 points each). Total: 60 points.

Part I follows the same structure as Examination 1 but draws from all ten weeks of content. Representative items from Part I are provided below; the remaining items follow the same format as those in Section B4.1.

Additional representative MCQ items — Weeks 6–10

- 21. In a Kaggle dataset documentation page, which element most directly informs an ethical assessment of data provenance?
 - a) The file size of the dataset.
 - b) The description of how and from whom the data were collected, including any consent information.
 - c) The number of upvotes the dataset has received.
 - d) The format of the column headers.
- 22. An AI system that generates synthetic images of individuals in professional settings without their consent raises concerns primarily related to:
 - a) Model accuracy b) Overfitting c) Autonomy and consent d) Transfer learning
- 23. Which of the following best characterises the concept of “AI access inequality” in the Latin American context?
 - a) Equal access to AI tools across all income groups.
 - b) Differential access to AI education and infrastructure based on socioeconomic status, geographic location, and language, resulting in unequal distributions of AI’s benefits and harms.
 - c) A policy restricting AI use to government agencies.
 - d) A uniform level of AI literacy across all university students in the region.

24. A student uses ChatGPT to draft a section of their course assignment and submits it without disclosure. From an academic integrity perspective, the primary concern is:
 - a) The model may have used copyrighted training data.
 - b) The student has not demonstrated their own learning and has misrepresented the work as their own.
 - c) The text may contain grammatical errors.
 - d) The model operates on a remote server.
25. An end-to-end AI pipeline typically includes, in order:
 - a) Deployment → Data collection → Model training → Evaluation.
 - b) Problem framing → Data collection and cleaning → Model training → Evaluation → Deployment.
 - c) Model selection → Data collection → Problem framing → Evaluation.
 - d) Data collection → Deployment → Evaluation → Model training.

Part II: Short-Answer Questions (3 items, 10 points each)

31. *A public health agency in a low-income country wishes to deploy an AI model to prioritise patients for scarce medical resources. Identify three ethical principles that should guide the design and deployment of this system, and for each principle, describe one concrete design choice that would operationalise it. (100–150 words)*
32. *Compare the instructional approach you experienced in this course with an alternative approach you are familiar with (e.g., from another course). In your comparison, identify one advantage and one limitation of each approach for developing AI literacy. (100–150 words)*
33. *Select one AI application from your own academic discipline or intended profession. Describe how it works at a conceptual level, identify one technical challenge it faces, and explain one ethical concern its deployment raises. Use terminology from the course consistently and accurately. (100–150 words)*

B4.3. Individual Homework Assignments

Four individual homework assignments were due at the end of Weeks 2, 4, 6, and 8. Each assignment required a written individual response of 300–500 words submitted via the institutional LMS. Students could consult class notes and the assigned readings; AI tool use was not prohibited but had to be disclosed. Assignments were graded by the control instructor using the marking criteria described below.

Homework 1 (due end of Week 2)

Prompt: Select one application of Artificial Intelligence that you encounter in your daily life (e.g., a social media recommendation feed, a voice assistant, a spam filter). In 300–500 words: (a) explain what type of machine learning the application likely uses and why; (b) describe one specific way in which the quality or characteristics of the training data would affect the system’s output; and (c) identify one question about this system that you cannot answer with the knowledge gained so far.

Marking criteria: Accuracy of the ML type identification (30%); clarity and specificity of the data-quality analysis (40%); quality of the open question, including appropriate scoping and reference to course concepts (30%).

Homework 2 (due end of Week 4)

Prompt: Using the dataset provided in Weeks 3–4 (available in the course LMS folder), run the pre-built Google Colab notebook and complete the following individually: (a) run all cells and confirm the notebook executes without errors; (b) add a markdown cell after the final visualisation explaining, in plain language, what the chart shows and what a decision-maker could conclude from it; and (c) write a 100-word reflection on one decision you would make differently if you were redesigning this analysis and why. Submit the link to your personal Colab copy via LMS.

Marking criteria: Notebook executes correctly (20%); accuracy and clarity of the markdown explanation (40%); depth and specificity of the methodological reflection (40%).

Homework 3 (due end of Week 6)

Prompt: Search Kaggle for a dataset related to your academic discipline or intended profession. In 300–500 words: (a) describe the dataset (source, number of records, variables, stated collection method); (b) formulate one research question that this dataset could help answer; (c) identify one potential ethical concern about the data collection or intended use of this dataset; and (d) briefly explain what type of analysis (e.g., classification, regression, clustering) would be appropriate for your research question. Include the Kaggle dataset URL.

Marking criteria: Completeness and accuracy of dataset description (25%); clarity of the research question (25%); depth of ethical analysis (30%); appropriateness of analytical approach (20%).

Homework 4 (due end of Week 8)

Prompt: Read the UNESCO (2023) guidance sections assigned in Week 7. In 300–500 words: (a) identify what you consider the most important recommendation from the document for university students in your context; (b) explain why you chose this recommendation, connecting it to at least one specific risk or opportunity you have observed or could plausibly observe in your own academic environment; and (c) describe one concrete change to your own behavior regarding AI tool use that you would make based on this recommendation.

Marking criteria: Quality of recommendation selection and justification (35%); specificity of the contextual connection (35%); concreteness and plausibility of the proposed behavioral change (30%).

B5. Treatment-Differentiation Verification

An independent observer attended two control-group sessions (Sessions 6 and 14, selected from different weeks and different hours of the day) to verify that no connectivist-framework components were introduced. The observer used the reverse side of the 15-indicator fidelity checklist (Supplementary Material D), recording the *absence* of the following elements in each observed session:

- Structured multidisciplinary team formation.
- Peer-teaching rotations.
- Structured ethical debate (Structured Academic Controversy).
- Mandatory GitHub commits.
- Digital portfolio entries.
- Network-diagram construction.
- Peer-assessment rubric application.

- Kaggle dataset exploration with collaborative task.

In both observed sessions, none of these elements was present. The control instructor delivered content through lecture and question-and-answer, consistent with the format described in Section B2. Students used ChatGPT incidentally (one student was observed querying the model independently during free time), but no structured activity requiring its use was assigned. The observer’s notes confirmed treatment differentiation for the observed sessions.

B6. Equivalence Assurances

The following measures ensured that observed outcome differences between conditions can be attributed to the pedagogical approach rather than to differences in content, dosage, or access:

- **Same content guide.** Both the experimental facilitators and the control instructor received an identical standardized content guide specifying topics, readings, and learning objectives week by week.
- **Same contact hours.** Both conditions totalled exactly 30 instructional hours across 20 sessions.
- **Same readings.** The readings listed in Table B2 are identical to those assigned in the experimental condition (Supplementary Material A, Table A3).
- **Same tool access.** Both conditions had access to ChatGPT and Google Colab; only the structure of tool use differed.
- **Same measurement schedule.** Both conditions completed T0 (Week 0), T1 (Weeks 8–12, per each section’s calendar), and T2 (four weeks post-intervention) under identical standardized conditions.
- **Separate personnel.** The control instructor had no contact with the experimental facilitators’ supervision meetings and received no training in the connectivist framework, minimising diffusion of treatment.
- **Waitlist access.** Upon study completion, control participants were offered full access to the experimental materials, addressing the ethical implications of differential treatment.